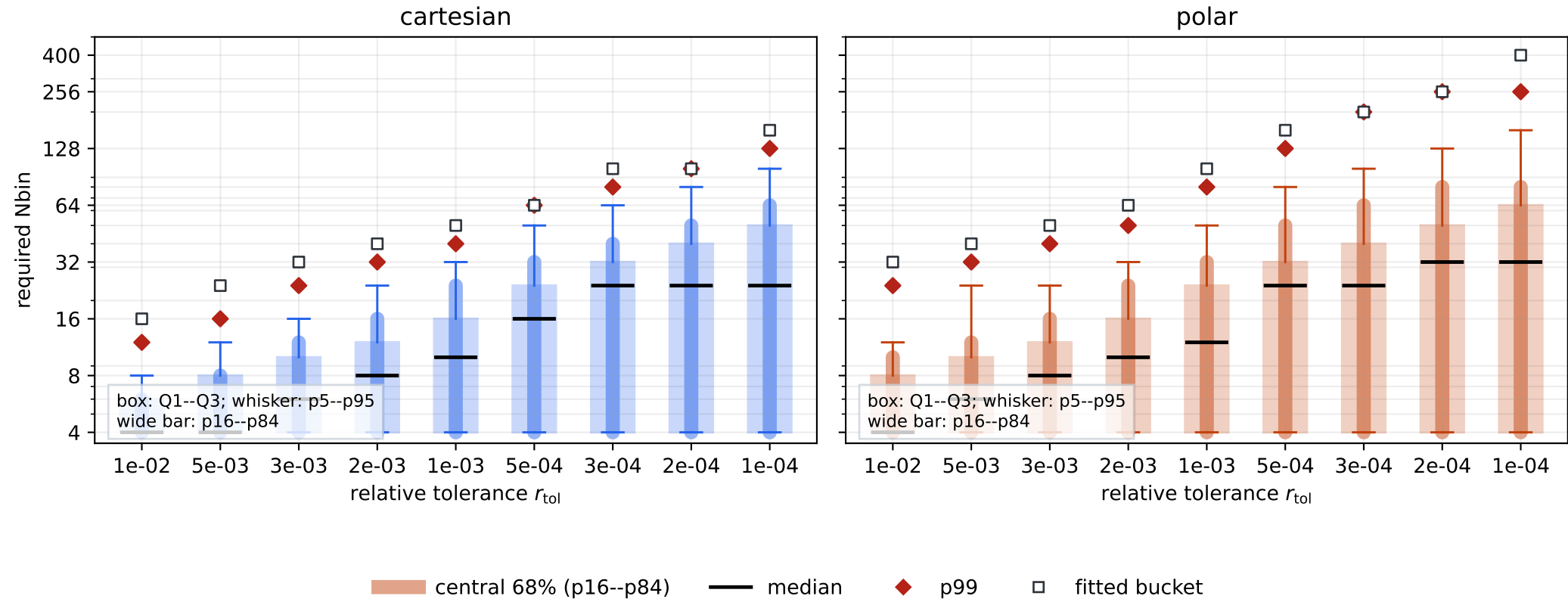
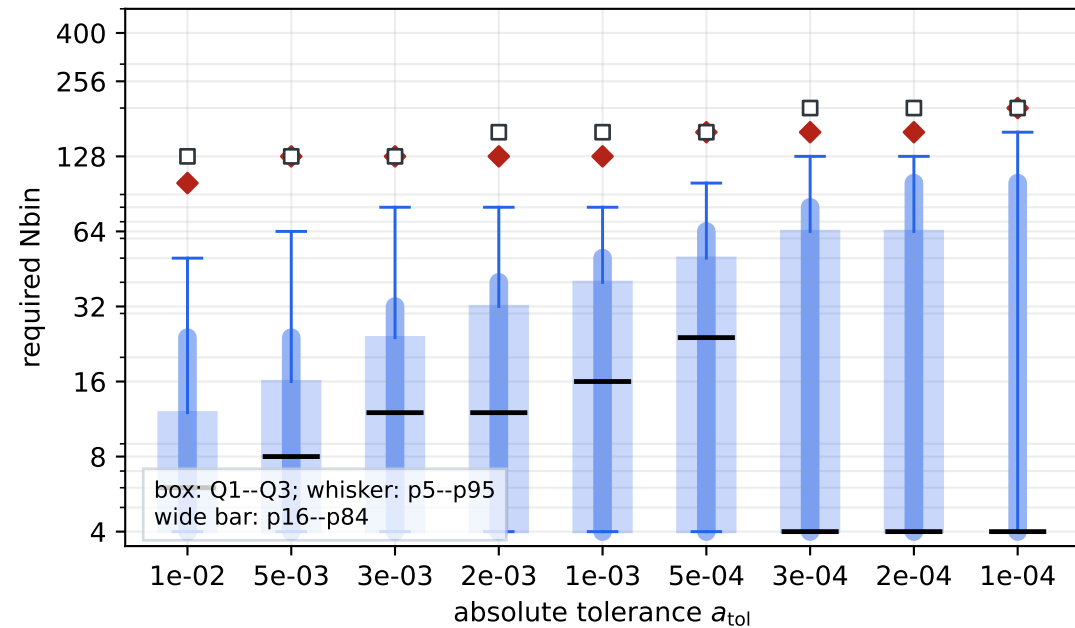


Relative branch: required resolution on the independent holdout

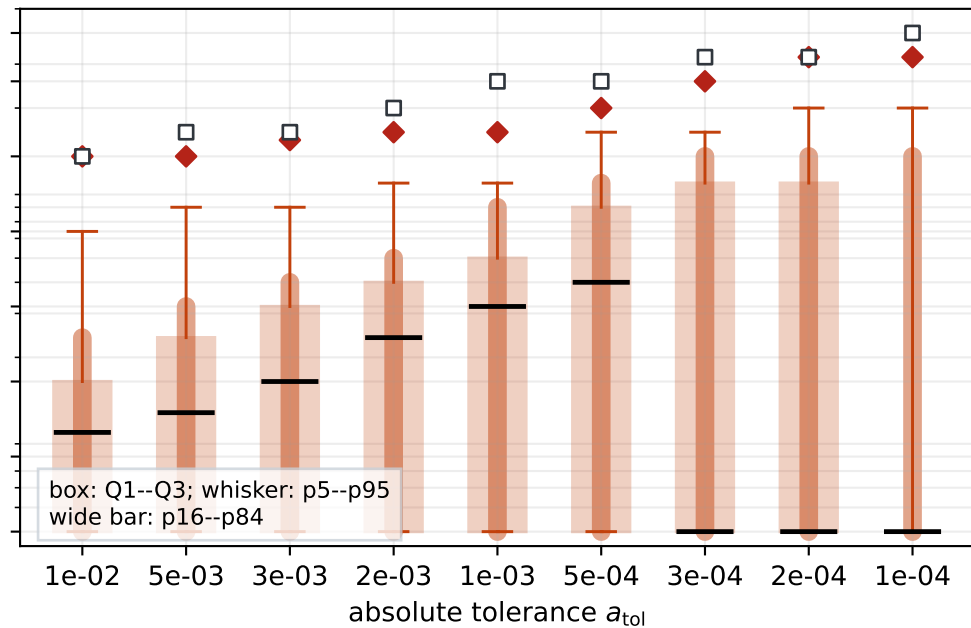


Absolute branch: required resolution on the independent holdout

cartesian



polar



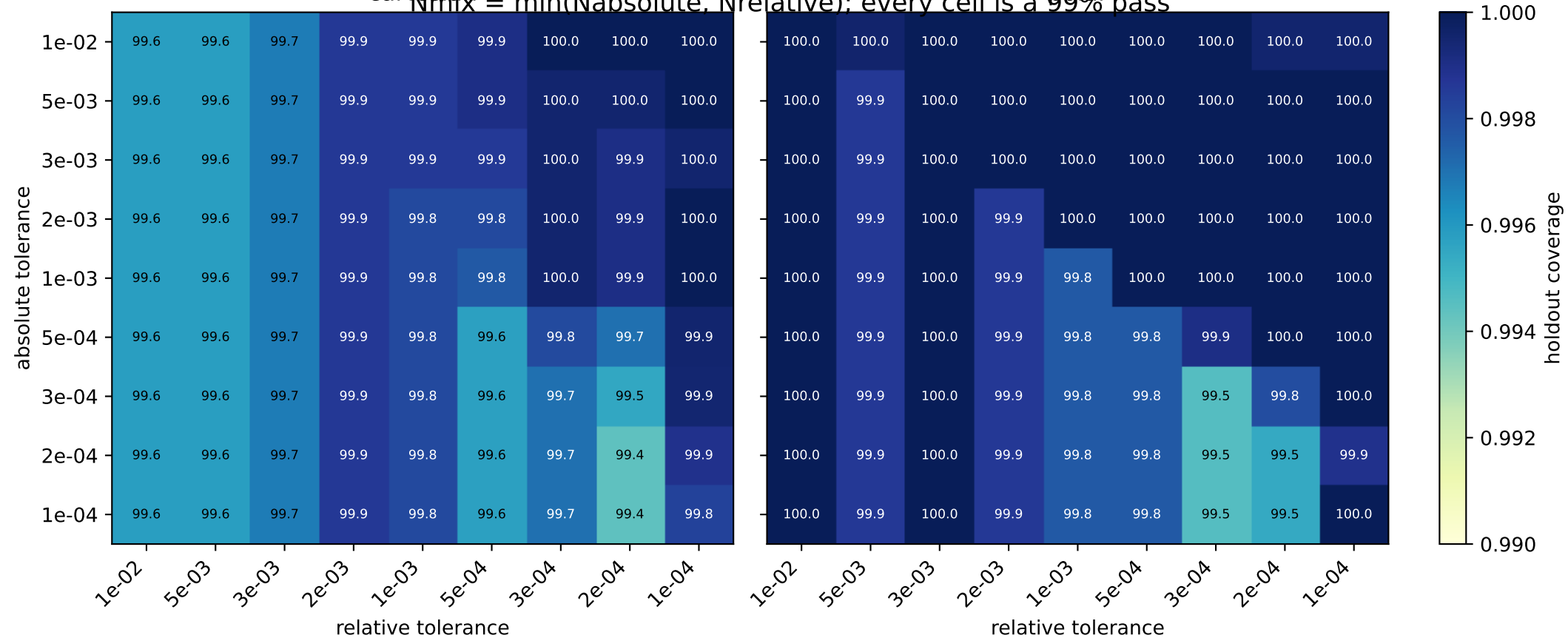
central 68% (p16--p84)

median

p99

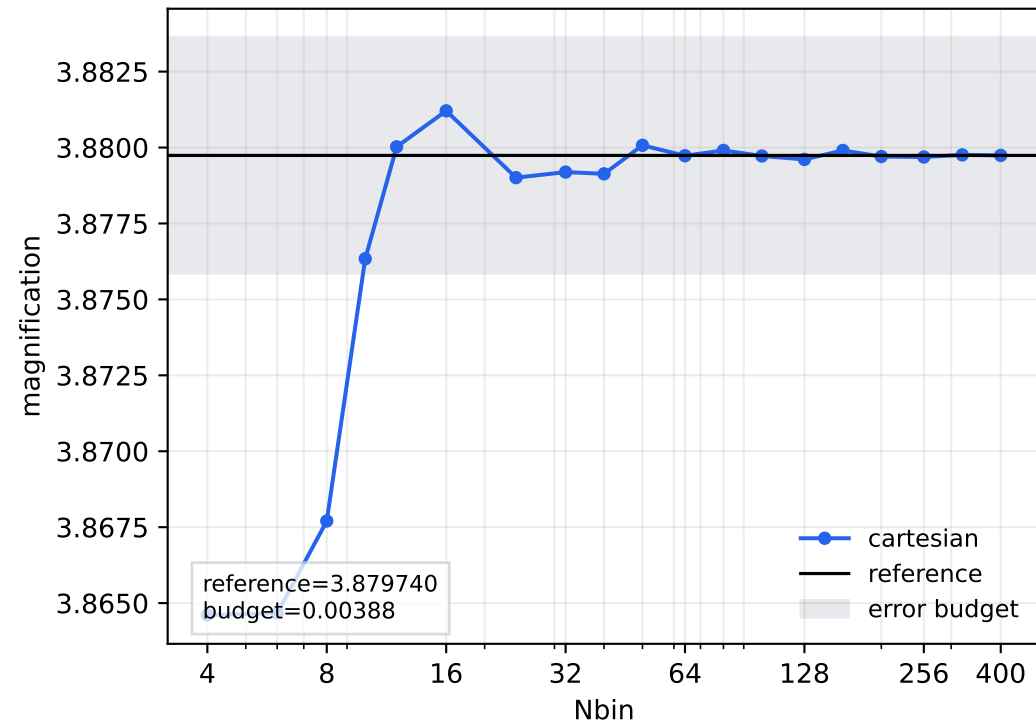
fitted bucket

Mixed max-budget rule: holdout coverage (%)
 $N_{mix} = \min(N_{absolute}, N_{relative})$; every cell is a 99% pass

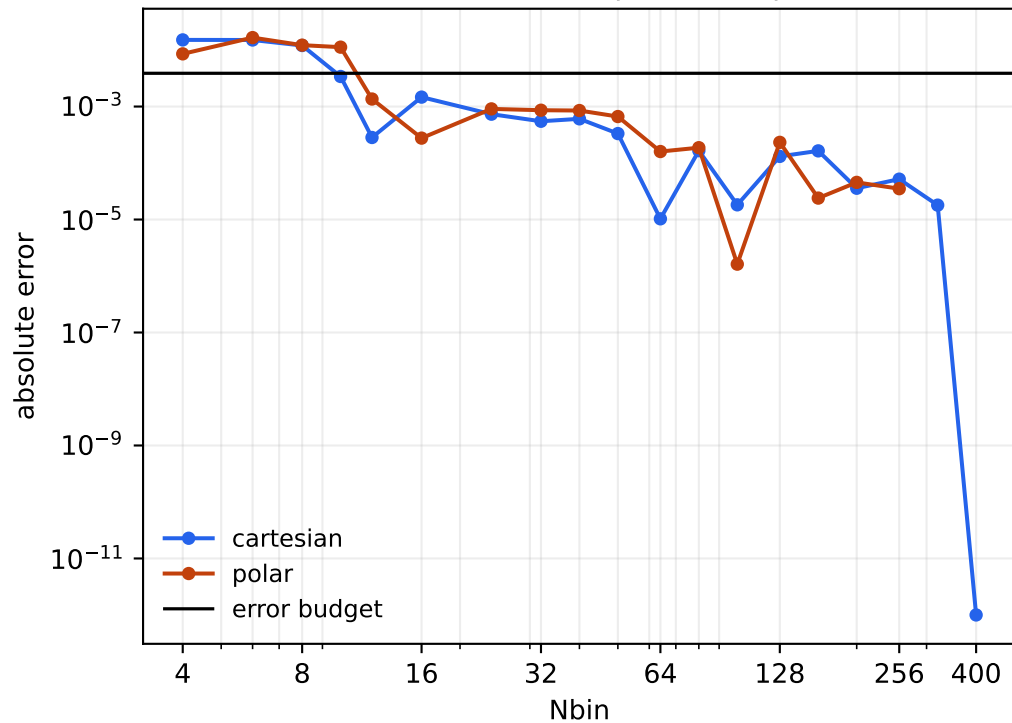


Representative holdout case: case 48 (linear)
reltol=1e-3; required buckets: Cartesian 10, polar 12

cartesian: A(N)



absolute error $|A(N)-A_{ref}|$



A common empirical rule for both tolerance branches

1. For every row, increase Nbin and record the first persistent crossing of the requested budget against the high-resolution reference.
2. Fit the discovery p99 in base-two logarithms, separately for the relative and absolute branches. The grid only changes C and beta.
3. At runtime policy level, use $B = \max(B_{abs}, B_{rel})$, hence $\epsilon = \max(atol / \max(|A|, 1), reltol)$ and $N = \min(N_{abs}, N_{rel})$.
4. The 9x9 mixed holdout matrix is a validation of this composition, not a second fit. All 162 cells clear the 99% target.

grid	Crel	beta-rel	Cabs	beta-abs	mixed min
cartesian	45.3	0.477	140.5	0.110	99.44%
polar	94.6	0.595	201.0	0.229	99.46%

The fitted laws are population-level p99 rules; the runtime estimator is a separate claim.
This PDF documents calibration and validation, not a production C++ selector change.